

Qubot-Publication V1.0.1

This release contains the code and collected data from the experiments presented in the publication:

“Open-Source Modular Laboratory Robots as Building Blocks for Flexible Assembly of Self-Driving Labs”

Contents

- **README.pdf**: general description and instructions to use repository files
- **data.zip**: contains raw and processed experiment datasets
- **qubot.zip**: contains latest release from the Qubo public repository (**mat-fox/qubot**: Qubot modular lab automation platform)
- **scripts.zip**: contains code used to analyze and visualize the datasets

Instructions

- Place the **data/** and **scripts/** folders in the same parent directory before running any analysis or visualization scripts.
- The analysis and visualization scripts expect the repository structure to include both **data/** and **scripts/** at the same level, so relative paths resolve correctly.
- **scripts/shampoo/formulation_selection.py** requires the Liquid Formulations Dataset file from the publication Chitre, A., et al. Sci Data 11, 728 (2024) (<https://doi.org/10.6084/m9.figshare.c.7132624>) to be downloaded to the **data/shampoo** directory as **LiquidFormulationsDataset_2023.json**.

Data folder structure

data/electrolytes

- Contains electrolyte experiment datasets and summaries
- **summary.csv**: aggregate summary for the electrolytes dataset
- **viscous_liquids_parameters.json**: parameters used for viscous liquids used during experimentation
- Individual experiment folders named by date and time, such as **01072025_1007** and **30062025_1415**
 - **Distributed_coincell.csv**: Aggregated data of the gravimetric transfer of samples into coin cells
 - **Summary_transfers.csv**: Aggregated data of the gravimetric transfer of reagents during formulation
 - **displacement/**: folder containing force measurements used for estimation of electrolyte thickness for each sample
 - **EIS/**: folder containig electrochemical impedance spectroscopy data for each sample

data/shampoo

- Contains shampoo experiment datasets and summaries
- **summary.csv**: aggregate summary for the shampoo dataset

- `stability.csv`: stability measurement results
- `viscous_liquids_parameters.json`: configuration parameters for shampoo reagents during experimentation
- Individual experiment folders named by date and time, such as `08122025_0856` and `26112025_0850`
 - `stability_images_24h/`: folder containing stability images after 24 hours
 - `pH_adjstment.csv`: files containing data during pH adjustment
 - `Summary_transfers.csv`: gravimetric testing of reagent transfers during experimentation

data/nominal_testing

Contains nominal testing datasets for different hardware platforms:

- Bending Test Data - Qubot Cantilever: CSV files containing data from bending tests of Qubot cantilever modules
- Repeatability and Least Increment Step Test Data - Ender 3 V2: CSV files containing nominal testing data for Ender 3 V2
- Repeatability and Least Increment Step Test Data - Genmitsu 3018 PROVer: CSV files containing nominal testing data for Genmitsu 3018 PROVer
- Repeatability and Least Increment Step Test Data - Opentrons OT-2: CSV files containing nominal testing data for Opentrons OT-2
- Repeatability and Least Increment Step Test Data - Qubot Gantry Aluminium: CSV files containing nominal testing data for Qubot Gantry with aluminium adapters
- Repeatability Test Data - Qubot Gantry PETG: CSV files containing nominal testing data for Qubot Gantry with PETG adapters

Qubot folder structure

- `qubot/CAD/`: CAD, STL and technical drawings files for gantry-style Qubot
- `qubot/Manuals/`: assembly manuals and BOM for gantry-style Qubot

Scripts folder structure

scripts/electrolytes

- `data_analysis.py`: analysis routines for electrolyte experiment data
- `visualization.py`: plotting and visualization utilities

scripts/nominal_testing

- `data_acquisition.py`: script to automate data acquisition from gauge when reading was implemented in parallel to the test script
- `least_increment_step_test_protocol_opentrons.py`: Opentrons step test script
- `least_increment_step_test_protocol.py`: step test script for non-Opentrons devices
- `repeatability_test_protocol.py`: repeatability test script for non-Opentrons devices
- `repeatability_test_protocol_opentrons.py`: Opentrons repeatability script

scripts/shampoo

- `data_analysis.py`: shampoo data analysis routines
- `formulation_selection.py`: formulation selection methods
- `visualization.py`: visualization script for shampoo experiments

scripts/tutorial

- `qubot_controllably_tutorial.py`: tutorial to control qubot modules using controllably package